

Differentiable functions – 7

Extrema and optimisation

Homework

1 сентября 2026 года

Theoretical minimum. Fermat's theorem and the second-order test; Lagrange multipliers with one and several constraints; existence of an extremum through compactness or coercivity; the tangent line trick and the SOS method; optimisation over a parameter; the Karush–Kuhn–Tucker conditions for inequality constraints (statement).

1. Find the maximum of $\sin A \sin B \sin C$ over the angles of a triangle, and the maximum of $\cos A \cos B \cos C$; in both cases justify the existence of the maximum.
2. Find the rectangle of maximal area inscribed in an ellipse, and the cylinder of maximal volume inscribed in a sphere.
3. Let $a_1, \dots, a_n > 0$ be fixed. Minimise $\sum a_i x_i$ subject to $\prod x_i = 1$, $x_i > 0$, and deduce the weighted AM–GM inequality.
4. Prove that among all closed plane curves of a given length the circle encloses the largest area, assuming the existence of an extremal curve (Steiner's argument); explain where existence is used.
5. Find $\max(x^2y + y^2z + z^2x)$ over $x, y, z \geq 0$ with $x + y + z = 1$, and explain why the maximum is not attained at a symmetric point.
6. Let f be continuous on $[0, 1]$ and $\int_0^1 f = 0$. Find $\max \int_0^1 f(x)g(x) dx$ over all g with $\|g\|_\infty \leq 1$, and describe the maximisers.
7. Prove that the point minimising the sum of distances to three given points of the plane (the Fermat point) is either a vertex with an angle at least 120° or the point seeing all three sides at 120° .

8. Let A be a symmetric matrix. Prove that the maximum of $\langle Ax, x \rangle$ on the unit sphere equals the largest eigenvalue, and derive the whole Courant–Fischer minimax principle from the multiplier rule.
9. Minimise $\int_0^1 (f'(x))^2 dx$ over all $f \in C^1[0, 1]$ with $f(0) = 0$, $f(1) = 1$, and solve the same problem with the extra constraint $\int_0^1 f = 0$.
10. State the Karush–Kuhn–Tucker conditions for a smooth problem with inequality constraints and verify them on the problem $\max x + y$ subject to $x^2 + y^2 \leq 1$, $x \geq 0$.